

Physics Cup 2026 Problem 4

1 Key Assumptions

For this problem, we have assumed that water is inviscid, incompressible and irrotational. There must also be no net flux out of the main cylindrical disk, so the flux out of the 6 holes sum to 0 as the container is sealed apart from these 6 holes. Since the height of the cylinder h is small, we can assume it has negligible z dependence and work in 2D instead. Notice that the partitioned case can be equated to the unpartitioned case through the method of images with the extra constraint that the holes symmetric across the partition must have the same flow rate.

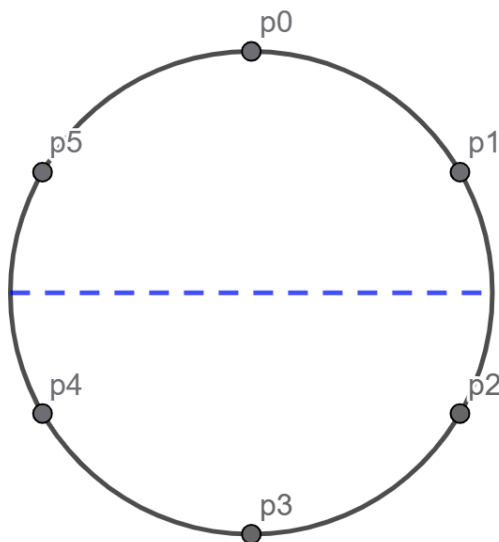


Figure 1: Diagram of the disk along with the indexing of the holes

2 Kinetic Energy Term

Since the flow is incompressible, inviscid and irrotational, a velocity potential

$$\mathbf{v} = \nabla\phi$$

can be defined. This potential satisfies laplace's equation in 2D, which has the general solution inside the disk

$$\phi(r, \theta) = \sum A_n r^{|n|} e^{in\theta} \quad (1)$$

Its normal derivative, which is with respect to r , at the circumference $r = R$ is

$$\partial_r \phi = \sum A_n |n| R^{|n|-1} e^{in\theta} \quad (2)$$

2.1 The Boundary Condition

We can set up a neumann boundary condition that is 0 at walls and q_i at across a hole. Since the arclength is very small, we will assume that there is a delta spike at angles that are multiples of $\pi/3$, so

$$g(\theta) = \frac{1}{Rd} \sum q_k \delta(\theta - \theta_k) \quad (3)$$

This can be written as a fourier series with coefficients

$$c_n = \frac{1}{2\pi} \int_0^{2\pi} g(\theta) e^{-in\theta} d\theta = \frac{1}{2\pi Rd} \sum q_k e^{-in\theta_k} \quad (4)$$

Note that $c_0 = 0$ as there will be no net flux out of the holes. Therefore, the neumann boundary condition (the normal derivative of ϕ) takes the form

$$g(\theta) = \frac{1}{2\pi Rd} \sum q_k e^{in(\theta - \theta_k)} \quad (5)$$

Equating this with Equation 2 and solving for A_n , we find that

$$\phi(R, \theta) = \sum \frac{c_n}{|n|} R e^{in\theta} \quad (6)$$

2.2 Finding Kinetic Energy

The kinetic energy of a system can be written as

$$T = \frac{\rho d}{2} \int |\nabla\phi|^2 dA \quad (7)$$

Green's First identity says that

$$\int \phi \nabla^2 \phi + \nabla\phi \cdot \nabla\phi dV = \oint \phi \nabla\phi \cdot d\mathbf{S}$$

And using laplace's equation, we can rewrite T as

$$T = \frac{\rho d}{2} \oint \phi g R d\theta = \pi \rho d R^2 \sum \frac{|c_n|^2}{|n|} = 2\pi \rho d R^2 \sum_{n=1} \frac{|c_n|^2}{n} \quad (8)$$

The last step is due to the fact that $c_n c_n^* = c_{-n} c_{-n}^*$. Focusing on the sum, we see that it can be expanded into

$$\sum_n \frac{1}{n} \sum_j \sum_k q_j q_k e^{in(\theta_j - \theta_k)}$$

Since we only care about the real part, this can be rewritten as

$$\sum_j \sum_k q_j q_k \sum_n \frac{\cos(n\Delta_{jk})}{n}, \quad \Delta_{jk} = \theta_j - \theta_k \quad (9)$$

It is a known formula that

$$\sum_{n=1} \frac{\cos(nx)}{n} = -\ln(2 \sin(x/2))$$

So kinetic energy can be written as

$$T = -\frac{\rho}{2\pi d} \sum_j \sum_k q_j q_k \ln(2 \sin(\Delta_{jk}/2)) \quad (10)$$

2.3 Cases of Δ_{jk}

There are 4 cases we have to consider for Δ_{jk} : when the difference $|j - k|$ is 0, 1, 2 or 3. This is because 4 gives the same result as 2; 5 with the same as 1. Evaluating these cases gives

$$T = -\frac{\rho}{2\pi d} (-\Lambda Q_0 + 2 \ln \sqrt{3} Q_2 + \ln 2 Q_3) \quad (11)$$

where

$$Q_i = \sum_m q_m q_{(m+i)\%6}$$

and % is the modulus sign. Λ is the logarithmically divergent term $\Lambda \sim \ln R/d$ that arises from approximating a finite sized hole using a dirac delta.

At last, we cannot forget about the water in the holes, which has total kinetic energy

$$T_h = \frac{\rho}{2d} Q_0$$

Therefore the total kinetic energy is

$$T = \frac{\rho}{2\pi d} ((\pi + \Lambda) Q_0 - \ln 3 Q_2 - \ln 2 Q_3) \quad (12)$$

3 Potential Energy Term

We will use Kinetic Energy Balance and Cauchy's Momentum Equation, which takes the form.

$$\rho \frac{D\mathbf{v}}{Dt} = \rho \mathbf{f} + \nabla \cdot \boldsymbol{\sigma} \quad (13)$$

Dotting both sides by $\mathbf{v}$ and integrating over a control volume, we notice that

$$\int \rho \mathbf{v} \cdot \frac{D\mathbf{v}}{Dt} dV = \frac{D}{Dt} \left(\int \frac{1}{2} \rho \mathbf{v}^2 dV \right) = \frac{DT}{Dt} = P \quad (14)$$

We will be taking the control volume to be made by all the water in the system, which is comoving with the liquid. This, by, Reynold's Transport Theorem, allows us to do this integration. Since there are no viscous stresses in this setup, the stress tensor reduces to $\boldsymbol{\sigma} = -p\mathbf{I}$, so

$$P = \int \rho \mathbf{v} \cdot \mathbf{f} dV - \oint p \mathbf{v} \cdot \mathbf{n} dS \quad (15)$$

Since the coriolis force does no work, and the centrifugal force can be written as a potential $\frac{1}{2}(\boldsymbol{\Omega} \times \mathbf{r})^2$, we can combine this with the pressure to form the reduced pressure and use the divergence theorem to remove the body force term:

$$p^* = p - \frac{1}{2} \rho (\boldsymbol{\Omega} \times \mathbf{r})^2$$

Since $\boldsymbol{\Omega}$ points along the $+z$ axis and $\mathbf{r}$ radially, so

$$p^* = p - \frac{1}{2} \rho \Omega^2 r^2$$

$$P = - \oint p^* \mathbf{v} \cdot \mathbf{n} dS \quad (16)$$

v_n is 0 against walls so the only place with nonzero v_n is at the menisci of the holes, where $v_n = q_k/d^2$. Therefore the total power can be written as

$$P = - \sum p_k^* q_k \quad (17)$$

The normal pressure p contains both a laplace jump and atmospheric pressure. The centrifugal radius is at $r = R + d + x_k$ as the water will have moved to that position already. Therefore

$$P = -p_0 \sum q_k - \sigma \sum \kappa_k q_k + \sum \frac{1}{2} \rho \Omega^2 (R^2 + 2R(d + x_k)) q_k \quad (18)$$

Where we have used the fact that $R \gg d \gg x_k$ to eliminate the quadratic terms.

For the meniscus we would like to assume that the contact line is mobile rather than fixed and the meniscus translates in the hole with negligible curvature. The problem specifies a 90° contact angle, which along with σ being big, suggests that any creation of curvature will be penalised so the meniscus will approximately remain flat and the leading-order pressure will remain 0. Hence, the laplace pressure term in can be neglected, and

$$P = \rho \Omega^2 R \sum x_k q_k \quad (19)$$

4 Final Equation of Motion

Since the entire system (including the partitioned case after being mapped to the unpartitioned case by method of images but with extra constraints on the symmetry of the modes) has C_6 symmetry, we can assume the water displacement takes the form $x_k = y_k \eta(t)$. where $y_k = y_0 \omega^k$ for the 6 possible modes of the system¹. x is actually a superposition of the 6 modes, but since we are only focused on the doubling time only the fastest growing mode will be taken. η acts as an evolution function of the holes. For convenience, we will perform a substitution $s_i = \frac{Q_i}{d^4 \eta^2}$. So the kinetic energy can be written as

$$T = \frac{1}{2} M \dot{\eta}^2, \quad M = \frac{\rho d^3}{\pi} ((\pi + \Lambda) s_0 - \ln 3 s_2 - \ln 2 s_3) \quad (20)$$

The power can be written as

$$P = \alpha d^2 s_0 \eta \dot{\eta}, \quad \alpha = \rho \Omega^2 R$$

Using $P = -\dot{U}$, the potential energy is

$$U = -\frac{1}{2} \alpha d^2 s_0 \eta^2 = -\frac{1}{2} K \eta^2, \quad K = \alpha d^2 s_0 \quad (21)$$

The Euler-Lagrange Equation gives

$$M \ddot{\eta} - K \eta = 0 \quad (22)$$

The most common anstaz for this differential equation is $\eta = e^{\lambda t}$, which gives us

$$\lambda^2 = \frac{K}{M} = \frac{\alpha \pi s_0}{\rho d} ((\pi + \Lambda) s_0 - \ln 3 s_2 - \ln 2 s_3)^{-1} \quad (23)$$

The doubling time $\tau = \frac{\ln 2}{\lambda}$, so

$$\tau^2 = (\ln 2)^2 \frac{\rho d}{\alpha \pi s_0} ((\pi + \Lambda) s_0 - \ln 3 s_2 - \ln 2 s_3) \quad (24)$$

4.1 The Eigenmodes

The 6 possible eigenvectors y_k are

$$\begin{aligned} y^{(0)} &= (1, 1, 1, 1, 1, 1) \\ y^{(1)} &= \left(1, \frac{1}{2}, -\frac{1}{2}, -1, -\frac{1}{2}, \frac{1}{2}\right) \\ y^{(2)} &= \left(1, -\frac{1}{2}, -\frac{1}{2}, 1, -\frac{1}{2}, -\frac{1}{2}\right) \end{aligned}$$

¹Apologies for the ambiguous notations, I am running out of variables to use.

$$\begin{aligned}
y^{(3)} &= (1, -1, 1, -1, 1, -1) \\
y^{(4)} &= \left(0, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, 0, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right) \\
y^{(5)} &= \left(0, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, 0, -\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)
\end{aligned}$$

Of these 6 modes, 0, 4, 5 are not plausible as in 0, there exists net outflux and for 4 and 5 the hole that travelled x (the y_0 hole) has not moved at all in addition to being degenerate cases of 1 and 2.²

Out of the other 3 modes, we want the one that minimises τ^2 while still satisfying the constraints on y_k (mainly for case 2).

For the unpartitioned case (case 1), the fastest growing mode is $y^{(3)}$, so

$$\tau_1^2 = (\ln 2)^2 \frac{\rho d}{\alpha \pi} ((\pi + \Lambda) - \ln 3 + \ln 2)$$

For the partitioned case, we need to enforce an extra constraint: $y_0 = y_3$, $y_1 = y_2$, $y_4 = y_5$. This leads to $y^{(2)}$ being the only possible mode, so

$$\tau_2^2 = (\ln 2)^2 \frac{\rho d}{\alpha \pi} ((\pi + \Lambda) + \frac{1}{2} \ln 3 - \ln 2)$$

Taking the difference and substituting in α , we get

$$\tau_2^2 - \tau_1^2 = (\ln 2)^2 \frac{d}{\pi \Omega^2 R} \ln \left(\frac{\sqrt{27}}{4} \right) \quad (25)$$

And indeed the logarithmically divergent term Λ cancels out.

²Indexing starts at 0 which is the target hole, as shown in Figure 1